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Electrochemistry

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MCQ (Single Correct Answer)	89 questions
Answer key	at the end

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MCQ (Single Correct Answer) 89 questions

Q1. RE-NEET 2026

The standard electrode potential (E°) for the half-cell reaction $Fe^{3+} + e^- \rightarrow Fe^{2+}$ at 298 K is (Given : $E^\circ\left(Fe^{3+}/Fe\right) = -0.04\ V$ and $E^\circ\left(Fe^{2+}/Fe\right) = -0.44\ V$ at 298 K)

(A) +0.92 V (B) +0.40 V (C) +0.76 V (D) -0.48 V

Open solution

Q2. RE-NEET 2026

For a salt XY, which is a strong electrolyte, the plot of Λ_m versus $\sqrt{c}$ has a slope of $-90.0\ S\ cm^2\ mol^{-3/2}L^{1/2}$ at 298 K . At 0.01 M concentration of $\mathbf{XY}$, the value of Λ_m is $145.0\ S\ cm^2\ mol^{-1}$. The limiting molar conductivity of $\mathbf{Y}^-$ ion $\left(\lambda^0_{Y^-}\right.$, in $\left.Scm^2\ mol^{-1}\right)$ at 298 K will be

(Given : $\lambda^0_{X^+} = 74.0\ S\ cm^2\ mol^{-1}$)

(A) 76.0 (B) 80.0 (C) 100.0 (D) 90.0

Open solution

Q3. NEET 2026

A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :

(Given : Molar mass of $Cu = 63\ g\ mol^{-1}$;

$1\ F = 96487C\ mol^{-1}$)

(A) 1.7018 g (B) 0.2938 g (C) 2.4036 g (D) 0.5876 g

Open solution

Q4. NEET 2026

Calculate emf of the half cell given below :

$Pt(s)\,|H_2(g, 2\ atm)|HCl(aq, 0.02M)$

$E^\circ_{H_2/H^+} = 0\ V$

(Given: $\frac{2.303RT}{F} = 0.059, \log 2 = 0.3010$)

(A) -0.109 V (B) 0.035 V (C) -0.035 V (D) 0.109 V

Open solution

Q5. NEET 2024 (Re-Examination)

From the following select the one which is not an example of corrosion.

(A) Rusting of iron object

(B) Production of hydrogen by electrolysis of water

(C) Tarnishing of silver

(D) Development of green coating on copper and bronze ornaments

Open solution

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Q6. NEET 2024 (Re-Examination)

The standard cell potential of the following cell $Zn\left|Zn^{2+}(aq)\right|Fe^{2+}(aq)\mid Fe$ is $0.32\ V$. Calculate the standard Gibbs energy change for the reaction:

$Zn(s)+Fe^{2+}(aq)\rightarrow Zn^{2+}(aq)+Fe(s)$

(Given : $1\ F=96487C$)

(A) $-61.75\ kJ\ mol^{-1}$

(B) $+5.006\ kJ\ mol^{-1}$

(C) $-5.006\ kJ\ mol^{-1}$

(D) $+61.75\ kJ\ mol^{-1}$

Open solution

Q7. NEET 2024

Match List I with List II.

	List I (Conversion)		List II (Number of Faraday required)
A.	1 mole of H_2O to O_2	I.	3F
B.	1 mol of MnO_4^- to Mn^{2+}	II.	2F
C.	1.5 mol of Ca from molten $CaCl_2$	III.	1F
D.	1 mol of FeO to Fe_2O_3	IV.	5F

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III

(B) A-III, B-IV, C-I, D-II

(C) A-II, B-III, C-I, D-IV

(D) A-III, B-IV, C-II, D-I

Open solution

Q8. NEET 2024

Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is (Given : Molar mass of $Cu: 63\ g\ mol^{-1}, 1\ F=96487C$)

(A) 3.15 g

(B) 0.315 g

(C) 31.5 g

(D) 0.0315 g

Open solution

Q9. NEET 2023 Manipur

The E^{Θ} values for

$Al^+/Al=+0.55\ V$ and $Tl^+/Tl=-0.34\ V$

$Al^{3+}/Al=-1.66\ V$ and $Tl^{3+}/Tl=+1.26\ V$

Identify the incorrect statement

(A) Al is more electropositive than Tl

(B) Tl^{3+} is a better reducing agent than Tl^{1+}

(C) Al^+ is unstable in solution

(D) Tl can be easily oxidised to Tl^+ than Tl^{3+}

Open solution

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Q10. NEET 2023 Manipur

Open solution

Molar conductance of an electrolyte increases with dilution according to the equation:

$\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$

Which of the following statements are true?

(A) This equation applies to both strong and weak electrolytes.

(B) Value of the constant A depends upon the nature of the solvent.

(C) Value of constant A is same for both $BaCl_2$ and $MgSO_4$

(D) Value of constant A is same for both $BaCl_2$ and $Mg(OH)_2$

Choose the most appropriate answer from the options given below:

(A) (A) and (B) only

(B) (A), (B) and (C) only

(C) (B) and (C) only

(D) (B) and (D) only

Q11. NEET 2023 Manipur

Open solution

The correct value of cell potential in volt for the reaction that occurs when the following two half cells are connected, is

$Fe^{2+}_{(aq)} + 2e^- \rightarrow Fe(s), E^\circ = -0.44\ V$

$Cr_2O_7^{2-}\ (aq) + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$E^\circ = +1.33\ V$

(A) $+1.77\ V$

(B) $+2.65\ V$

(C) $+0.01\ V$

(D) $+0.89\ V$

Q12. NEET 2023

Open solution

The conductivity of centimolar solution of KCl at $25^\circ C$ is $0.0210\ ohm^{-1}\ cm^{-1}$ and the resistance of the cell containing the solution at $25^\circ C$ is $60\ ohm$. The value of cell constant is -

(A) $3.28\ cm^{-1}$

(B) $1.26\ cm^{-1}$

(C) $3.34\ cm^{-1}$

(D) $1.34\ cm^{-1}$

Q13. NEET 2023

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A : In equation $\Delta_{rG} = -nFE_{cell}$, value of Δ_{rG} depends on n.

Reason R : E_{cell} is an intensive property and Δ_{rG} is an extensive property.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are true but R is NOT the correct explanation of A.

(B) A is true but R is false.

(C) A is false but R is true.

(D) Both A and R are true and R is the correct explanation of A.

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Q14. NEET 2022 Phase 2

Open solution

Two half cell reactions are given below.

$Co^{3+} + e^- \rightarrow Co^{2+}, \quad E^0_{Co^{2+}/Co^{3+}} = -1.81\,V$

$2Al^{3+} + 6e^- \rightarrow 2Al(s), \quad E^0_{Al/Al^{3+}} = +1.66\,V$

The standard EMF of a cell with feasible redox reaction will be :

(A) −3.47 V (B) +7.09 V (C) +0.15 V (D) +3.47 V

Q15. NEET 2022 Phase 2

Open solution

Standard electrode potential for the cell with cell reaction

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

is 1.1 V. Calculate the standard Gibbs energy change for the cell reaction. (Given F = 96487 C mol $^{-1}$)

(A) -200.27 J mol^{-1} (B) -200.27 kJ mol^{-1}

(C) -212.27 kJ mol^{-1} (D) -212.27 J mol^{-1}

Q16. NEET 2022 Phase 1

Open solution

At 298 K, the standard electrode potentials of Cu^{2+} / Cu, Zn^{2+} / Zn, Fe^{2+} / Fe and Ag^+ / Ag are 0.34 V, −0.76 V, −0.44 V and 0.80 V, respectively.

On the basis of standard electrode potential, predict which of the following reaction cannot occur?

(A) $CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$

(B) $CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)$

(C) $FeSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Fe(s)$

(D) $2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu(s) + Ag_2SO_4(aq)$

Q17. NEET 2022 Phase 1

Open solution

Given below are half cell reactions:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O,$

$E^o_{Mn^{2+}/MnO_4^-} = -1.510\,V$

$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$

$E^o_{O_2/H_2O} = +1.223\,V$

Will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid?

(A) Yes, because E^o_{cell} = + 0.287 V (B) No, because E^o_{cell} = −0.287 V

(C) Yes, because E^o_{cell} = + 2.733 V (D) No, because E^o_{cell} = − 2.733 V

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Q18. NEET 2022 Phase 1

Find the emf of the cell in which the following reaction takes place at 298 K

$Ni(s) + 2Ag^{+}$ (0.001 M) $\rightarrow Ni^{2+}$ (0.001 M) + 2Ag(s)

(Given that E^{o}_{cell} = 10.5 V, $\frac{2.303\ RT}{F} = 0.059$ at 298 K)

(A) 10.4115 V (B) 10.385 V (C) 0.9615 V (D) 10.05 V

Open solution

Q19. NEET 2021

The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and 91.0 S cm^2 mol^{-1} respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.

(A) 540.48 S cm^2 mol^{-1} (B) 201.28 S cm^2 mol^{-1}

(C) 390.71 S cm^2 mol^{-1} (D) 698.28 S cm^2 mol^{-1}

Open solution

Q20. NEET 2021

The molar conductivity of 0.007 M acetic acid is 20 S cm^2 mol^{-1}. What is the dissociation constant of acetic acid? Choose the correct option.

[$\Lambda^{o}_{H^+}$ = 350 S cm^2 mol^{-1}

$\Lambda^{o}_{CH_3COO^-}$ = 50 S cm^2 mol^{-1}]

(A) 2.50×10^{-5} mol L^{-1} (B) 1.75×10^{-4} mol L^{-1}

(C) 2.50×10^{-4} mol L^{-1} (D) 1.75×10^{-5} mol L^{-1}

Open solution

Q21. NEET 2020 Phase 2

In a typical fuel cell, the reactants (R) and product (P) are

(A) $R = H_{2(g)}, N_{2(g)}; P = NH_{3(aq)}$ (B) $R = H_{2(g)}, O_{2(g)}; P = H_2O_{2(\ell)}$

(C) $R = H_{2(g)}, O_{2(g)}; P = H_2O_{(\ell)}$ (D) $R = H_{2(g)}, O_{2(g)}, Cl_{2(g)}; P = HClO_{4(aq)}$

Open solution

Q22. NEET 2020 Phase 2

Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at 298 K.

(A) $K^{+} + 1e^{-} \rightarrow K_{(s)}$ (B) $Mg^{2+} + 2e^{-} \rightarrow Mg_{(s)}$

(C) $Fe^{2+} + 2e^{-} \rightarrow Fe_{(s)}$ (D) $Au^{3+} + 3e^{-} \rightarrow Au_{(s)}$

Open solution

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Q23. NEET 2020 Phase 1

On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :

(A) Oxygen gas (B) H_2S gas (C) SO_2 gas (D) Hydrogen gas

Open solution

Q24. NEET 2020 Phase 1

The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is :

(A) 2 (B) 3 (C) 4 (D) 1

Open solution

Q25. NEET 2019 (Odisha)

Following limiting molar conductivities are given as

$\lambda_m^\circ(H_2SO_4) = x\ S\ cm^2\ mol^{-1}$

$\lambda_m^\circ(K_2SO_4) = y\ S\ cm^2\ mol^{-1}$

$\lambda_m^\circ(CH_3COOK) = z\ S\ cm^2\ mol^{-1}$

λ_m° (in $S\ cm^2\ mol^{-1}$) for CH_3COOH will be

(A) $\frac{(x-y)}{2} + z$ (B) $x - y + 2z$ (C) $x + y + z$ (D) $x - y + z$

Open solution

Q26. NEET 2019 (Odisha)

The standard electrode potential ($E^\ominus$) values of Al^{3+}/Al, Ag^+/Ag, K^+/K and Cr^{3+}/Cr are -1.66 V, 0.80 V, -2.93 V and -0.74 V, respectively. The correct decreasing order of reducing power of the metal is

(A) $Al > K > Ag > Cr$ (B) $Ag > Cr > Al > K$

(C) $K > Al > Cr > Ag$ (D) $K > Al > Ag > Cr$

Open solution

Q27. NEET 2019

For a cell involving one electron E^Θ_{cell} = 0.59 V at 298 K, the equilibrium constant for the cell reaction is :

[Given that $\frac{2.303RT}{F}$ = 0.059 V at T = 298 K]

(A) 1.0×10^{30} (B) 1.0×10^{10} (C) 1.0×10^{2} (D) 1.0×10^{5}

Open solution

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Q28. NEET 2019

For the cell reaction
$2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$
E^{Θ}_{cell} = 0.24 V at 298 K. The standard Gibbs energy ($\Delta_r G^o$) of the cell reaction is :
[Given that Faraday constant F = 96500 C mol^{-1}]

(A) 46.32 kJ mol^{-1} (B) 23.16 kJ mol^{-1} (C) −46.32 kJ mol^{-1} (D) −23.16 kJ mol^{-1}

Open solution

Q29. NEET 2018

Consider the change in oxidation state of bromine corresponding to different emf values as shown in the given diagram :

$$BrO_4^- \xrightarrow{1.82\ V} BrO_3^- \xrightarrow{1.5\ V} HBrO \xrightarrow{1.595\ V} Br_2 \xrightarrow{1.0652\ V} Br^-$$

Then the species undergoing disproportionation is :

(A) BrO_4^- (B) BrO_3^- (C) Br_2 (D) HBrO

Open solution

Q30. NEET 2017

In the electrochemical cell :
$Zn\,|ZnSO_4\,(0.01\,M)||CuSO_4\,(1.0M)|Cu,$
the emf of this Daniell cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0 M and that of CuSO4 changed to 0.01 M, the emf changes to E_2. From the followings, which one is the relationship between E_1 and E_2? (Given, RT/F = 0.059)

(A) $E_1 < E_2$ (B) $E_1 > E_2$ (C) $E_2 = 0^1E_1$ (D) $E_1 = E_2$

Open solution

Q31. NEET 2016 Phase 2

The molar conductivity of a 0.5 mol/dm^3 solution of $AgNO_3$ with electrolytic conductivity of 5.76 $\times$ 10^{-3} S cm^{-1} at 298 K is

(A) 2.88 S cm^2/mol (B) 11.52 S cm^2/mol (C) 0.086 S cm^2/mol (D) 28.8 S cm^2/mol

Open solution

Q32. NEET 2016 Phase 2

During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is

(A) 55 minutes (B) 110 minutes (C) 220 minutes (D) 330 minutes

Open solution

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Q33. NEET 2016 Phase 2

If the E^o_{cell} for a given reaction has a negative value, which of the following gives the correct relationships for the values of ΔG^o and K_{eq} ?

(A) $\Delta G^o > 0;\ K_{eq} < 1$

(B) $\Delta G^o > 0;\ K_{eq} > 1$

(C) $\Delta G^o < 0;\ K_{eq} > 1$

(D) $\Delta G^o < 0;\ K_{eq} < 1$

Open solution

Q34. NEET 2016 Phase 2

The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds is (charge on electron = 1.60×10^{-19}C)

(A) 6×10^{23}

(B) 6×10^{20}

(C) 3.75×10^{20}

(D) 7.48×10^{23}

Open solution

Q35. NEET 2016 Phase 2

Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because

(A) zinc is lighter than iron

(B) zinc has lower melting point than iron

(C) zinc has lower negative electrode potential than iron

(D) zinc has higher negative electrode potential than iron

Open solution

Q36. NEET 2016 Phase 1

The pressure of H_2 required to make the potential of H_2-electrode zero in pure water at 298 K is

(A) 10^{-10} atm

(B) 10^{-4} atm

(C) 10^{-14} atm

(D) 10^{-12} atm

Open solution

Q37. AIPMT 2015

Aqueous solution of which of the following compounds is the best conductor of electric current ?

(A) Hydrochloric acid, HCl

(B) Ammonia, NH_3

(C) Fructose, $C_6H_{12}O_6$

(D) Acetic acid, $C_2H_4O_2$

Open solution

Q38. AIPMT 2015 Cancelled Paper

A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as

(A) dynamo

(B) Ni-Cd cell

(C) fuel cell

(D) electrolytic cell

Open solution

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Q39. AIPMT 2014

When 0.1 mol MnO_4^{2-} is oxidised the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is

(A) 96500 C (B) 2×96500 C (C) 9650 C (D) 96.50 C

Open solution

Q40. AIPMT 2014

The weight of silver (at. wt. = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be

(A) 5.4 g (B) 10.8 g (C) 54.0 g (D) 108.0 g

Open solution

Q41. NEET 2013 (Karnataka)

Consider the half-cell reduction reaction

$Mn^{2+} + 2e^- \rightarrow Mn$, $E^o = -1.18$ V

$Mn^{2+} \rightarrow Mn^{3+} + e^-$, $E^o = -1.51$ V

The E^o for the reaction $3\ Mn^{2+} \rightarrow Mn^o + 2Mn^{3+}$, and possibility of the forward reaction are respectively

(A) $-$ 4.18 V and yes (B) + 0.33 V and yes (C) + 2.69 V and no (D) $-$ 2.69 V and no

Open solution

Q42. NEET 2013 (Karnataka)

How many gram of cobalt metal will be deposited when a solution of cobalt (II) chloride is electrolyzed with a current of 10 amperes for 109 minutes (1 Faraday = 96,500 C; Atomic mass of Co = 59 u)

(A) 4.0 (B) 20.0 (C) 40.0 (D) 0.66

Open solution

Q43. NEET 2013

At 25°C molar conductance of 0.1 molar aqueous solution of ammonium hydroxide is 9.54 $ohm^{-1}\ cm^2\ mol^{-1}$ and at infinite dilution its molar conductance is 238 $ohm^{-1}\ cm^2\ mol^{-1}$. The degree of ionisation of ammonium hydroxide at the same concentration and temperature is

(A) 4.008% (B) 40.800% (C) 2.080% (D) 20.800%

Open solution

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Q44. NEET 2013

A button cell used in watches function as following.

$Zn_{(s)} + Ag_2O_{(s)} + H_2O_{(l)} \rightleftharpoons 2Ag_{(s)} + Zn^{2+}_{(aq)} + 2OH^-_{(aq)}$

If half cell potentials are

$Zn^{2+}_{(aq)} + 2e^- \rightarrow Zn_{(s)}$; $E^o = -0.76$ V

$Ag_2O_{(s)} + H_2O_{(l)} + 2e^- \rightarrow 2Ag_{(s)} + 2OH^-_{(aq)}$, $E^o = 0.34$ V

The cell potential will be

(A) 0.84 V (B) 1.34 V (C) 1.10 V (D) 0.42 V

Open solution

Q45. NEET 2013

A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be

(A) 0.118 V (B) 1.18 V (C) 0.059 V (D) 0.59 V

Open solution

Q46. AIPMT 2012 Mains

Standard reduction potentials of the half reactions are given below :

$F_{2(g)} + 2e^- \rightarrow 2F^-_{(aq)}$; $E^o = +2.85$ V

$Cl_{2(g)} + 2e^- \rightarrow 2Cl^-_{(aq)}$; $E^o = +1.36$ V

$Br_{2(l)} + 2e^- \rightarrow 2Br^-_{(aq)}$; $E^o = +1.06$ V

$I_{2(s)} + 2e^- \rightarrow 2I^-_{(aq)}$; $E^o = +0.53$ V

The strongest oxidising and reducing agents respectively are

(A) F_2 and I^- (B) Br_2 and Cl^- (C) Cl_2 and Br^- (D) Cl_2 and I_2

Open solution

Q47. AIPMT 2012 Mains

Molar conductivities (Λ^o_m) at infinite dilution of NaCl, Hcl and CH_3COONa are 126.4, 425.9 and 91.0 S cm^2 mol^{-1} respectively. (Λ^o_m) for CH_3COOH will be

(A) 425.5 S cm^2 mol^{-1} (B) 180.5 S cm^2 mol^{-1}

(C) 290.8 S cm^2 mol^{-1} (D) 390.5 S cm^2 mol^{-1}

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Q48. AIPMT 2012 Mains

The Gibb's energy for the decomposition of Al_2O_3 at 500°C is as follows

$\frac{2}{3}\ Al_2O_3 \rightarrow \frac{4}{3}\ Al + O_2$

$\Delta_r G$ = +960 kJ mol^{-1}

The potential difference needed for the electrolytic reduction of aluminium oxide (Al_2O_3) at 500°C is at least

(A) 4.5 V (B) 3.0 V (C) 2.5 V (D) 5.0 V

Open solution

Q49. AIPMT 2012 Prelims

Limiting molar conductivity of NH_4OH [i.e. $\Lambda^0_{m(NH_4OH)}$] is equal to

(A) $\Lambda^0_{m(NH_4OH)} + \Lambda^0_{m(NaCl)} - \Lambda^0_{m(NaOH)}$

(B) $\Lambda^0_{m(NaOH)} + \Lambda^0_{m(NaCl)} - \Lambda^0_{m(NH_4Cl)}$

(C) $\Lambda^0_{m(NH_4OH)} + \Lambda^0_{m(NH_4Cl)} - \Lambda^0_{m(HCl)}$

(D) $\Lambda^0_{m(NH_4Cl)} + \Lambda^0_{m(NaOH)} - \Lambda^0_{m(NaCl)}$

Open solution

Q50. AIPMT 2011 Mains

A solution contains Fe^{2+}, Fe^{3+} and I^- ions. This solution was treated with iodine at 35°C. E^o for Fe^{3+}/Fe^{2+} is + 0.77 V and E^o for $I_2/2I^-$ = 0.536 V.

The favourable redox reaction is

(A) I_2 will be reduced to I^- (B) there will be no redox reaction

(C) I^- will be oxidised to I_2 (D) Fe^{2+} will be oxidised to Fe^{3+}

Open solution

Q51. AIPMT 2011 Prelims

Standard electrode potential of three metals X, Y and Z are −1.2 V, + 0.5 V and − 3.0 V respectively. The reducing power of these metals will be

(A) Y > Z > X (B) Y > X > Z (C) Z > X > Y (D) X > Y > Z

Open solution

Q52. AIPMT 2011 Prelims

The electrode potentials for $Cu^{2+}_{(aq)} + e^- \rightarrow Cu^+_{(aq)}$ and $Cu^+_{(aq)} + e^- \rightarrow Cu_{(s)}$ are + 0.15 V and + 0.50 V respectively.

The value of $E^o_{cu^{2+}/cu}$ will be

(A) 0.500 V (B) 0.325 V (C) 0.650 V (D) 0.150 V

Open solution

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Q53. AIPMT 2011 Prelims

Standard electrode potential for Sn^{4+}/Sn^{2+} couple is + 0.15 V and that for the Cr^{3+}/Cr couple is − 0.74 V. These two couples in their standard state are connected to make a cell. The cell potential will be

Open solution

(A) + 1.19 V (B) + 0.89 V (C) + 0.18 V (D) + 1.83 V

Q54. AIPMT 2010 Mains

Which of the following expressions correctly represents the equivalent conductance at infinite dilution of $Al_2(SO_4)_3$. Given that $\overset{\circ}{\Lambda}_{Al^{3+}}$ and $\overset{\circ}{\Lambda}_{SO_4^{2-}}$ are the equivalent conductances at infinite dilution of the respective ions?

Open solution

(A) $2\overset{\circ}{\Lambda}_{Al^{3+}} + 3\overset{\circ}{\Lambda}_{SO_4^{2-}}$

(B) $\overset{\circ}{\Lambda}_{Al^{3+}} + \overset{\circ}{\Lambda}_{SO_4^{2-}}$

(C) $(\overset{\circ}{\Lambda}_{Al^{3+}} + \overset{\circ}{\Lambda}_{SO_4^{2-}}) \times 6$

(D) $\frac{1}{3}\overset{\circ}{\Lambda}_{Al^{3+}} + \frac{1}{2}\overset{\circ}{\Lambda}_{SO_4^{2-}}$

Q55. AIPMT 2010 Mains

Consider the following relations for emf of an electrochemical cell

(i) EMF of cell = (Oxidation potential of anode) − (Reduction potential of cathode)
(ii) EMF of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
(iii) EMF of cell = (Reductional potential of anode) + (Reduction potential of cathode)
(iv) EMF of cell = (Oxidation potential of anode) − (Oxidation potential of cathode)

Which of the above relations are correct?

Open solution

(A) (iii) and (i) (B) (i) and (ii) (C) (iii) and (iv) (D) (ii) and (iv)

Q56. AIPMT 2010 Prelims

For the reduction of silver ions with copper metal, the standard cell potential was found to be + 0.46 V at 25°C. The value of standard Gibb's energy, ΔG^o will be
(F = 96500 C mol^{-1})

Open solution

(A) − 89.0 kJ (B) − 89.0 J (C) − 44.5 kJ (D) − 98.0 kJ

Q57. AIPMT 2010 Prelims

An increase in equivalent conductance of a strong electrolyte with dilution is mainly due to

Open solution

(A) increase in ionic mobility of ions

(B) 100% ionisation of electrolyte at normal dilution

(C) increase in both i.e., number of ions and ionic mobility of ions

(D) increase in number of ions.

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Q58. AIPMT 2009

Open solution

Given :

(i) $Cu^{2+} + 2e^- \rightarrow Cu,\ E^o = 0.337\ V$

(ii) $Cu^{2+} + e^- \rightarrow Cu^+,\ E^o = 0.153\ V$

Electrode potential, E^o for the reaction,

$Cu^+ + e^- \rightarrow Cu,$ will be

(A) 0.90 V (B) 0.30 V (C) 0.38 V (D) 0.52 V

Q59. AIPMT 2009

Open solution

Al_2O_3 is reduced by electrolysis at low potentials and high currents. If 4.0×10^4 amperes of current is passed through molten Al_2O_3 for 6 hours, what mass of aluminium is produced? (Assume 100% current efficiency, at. mass of Al = 27 g mol^{-1}).

(A) 8.1×10^4 g (B) 2.4×10^5 g (C) 1.3×10^4 g (D) 9.0×10^3 g

Q60. AIPMT 2009

Open solution

The equivalent conductance of M/32 solution of a weak monobasic acid is 8.0 mho cm^2 and at infinite dilution is 400 mho cm^2. The dissociation constant of this acid is

(A) 1.25×10^{-6} (B) 6.25×10^{-4} (C) 1.25×10^{-4} (D) 1.25×10^{-5}

Q61. AIPMT 2008

Open solution

On the basis of the following E^o values, the strongest oxidizing agent is

$[Fe(CN)_6]^{4-} \rightarrow [Fe(CN)_6]^{3-} + e^-;\ E^o = -0.35\ V$

$Fe^{2+} \rightarrow Fe^{3+} + e^-;\ E^o = -0.77\ V$

(A) Fe^{3+} (B) $[Fe(CN)_6]^{3-}$ (C) $[Fe(CN)_6]^{4-}$ (D) Fe^{2+}

Q62. AIPMT 2008

Open solution

Kohlrausch's law states that at

(A) Infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte

(B) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte

(C) Finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte

(D) Infinite dilution each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of the electrolyte.

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Q63. AIPMT 2008

Standard free energies of formation (in kJ/mol) at 298 K are -237.2, -394.4 and -8.2 for $H_2O_{(l)}$, $CO_{2(g)}$ and pentane $_{(g)}$ respectively. The value of E^o_{cell} for the pentane-oxygen fuel cell is

(A) 1.0968 V (B) 0.0968 V (C) 1.968 V (D) 2.0968 V

Open solution

Q64. AIPMT 2007

The equilibrium constant of the reaction:

$Cu_{(s)} + 2Ag^+_{(aq)} \rightarrow Cu^{2+}_{(aq)} + 2Ag_{(s)}$;

$E^o = 0.46$ V at 298 K is

(A) 2.0×10^{10} (B) 4.0×10^{10} (C) 4.0×10^{15} (D) 2.4×10^{10}

Open solution

Q65. AIPMT 2007

The efficiency of a fuel cell is given by

(A) $\Delta G/\Delta S$ (B) $\Delta G/\Delta H$ (C) $\Delta S/\Delta G$ (D) $\Delta H/\Delta G$

Open solution

Q66. AIPMT 2006

A hypothetical electrochemical cell is shown below.

$$A\,|A^+\,(xM)||B^+\,(yM)|B$$

The emf measured is + 0.20 V. The cell reaction is

(A) $A + B^+ \rightarrow A^+ + B$ (B) $A^+ + B \rightarrow A + B^+$

(C) $A^+ + e^- \rightarrow A$; $B^+ + e^- \rightarrow B$ (D) the cell reaction cannot be predicted.

Open solution

Q67. AIPMT 2006

$E^o_{Fe^{2+}/Fe} = -0.441$ V and $E^o_{Fe^{3+}/Fe^{2+}} = 0.771$ V, the standard EMF of the reaction $Fe + 2Fe^{3+} \rightarrow 3Fe^{2+}$ will be

(A) 0.111 V (B) 0.330 V (C) 1.653 V (D) 1.212 V

Open solution

Q68. AIPMT 2005

4.5 g of aluminium (at. mass 27 amu) is deposited at cathode from Al^{3+} solution by a certain quantity of electric charge. The volume of hydrogen produced at STP from H ions in solution by the same quantity of electric charge will be

(A) 44.8 L (B) 22.4 L (C) 11.2 L (D) 5.6 L

Open solution

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Q69. AIPMT 2005

The mass of carbon anode consumed (giving only carbon dioxide) in the production of 270 kg of aluminium metal from bauxite by the Hall process is
(Atomic mass : Al = 27)

Open solution

(A) 270 kg (B) 540 kg. (C) 90 kg (D) 180 kg

Q70. AIPMT 2004

The standard e.m.f. of a galvanic cell involving cell reaction with n = 2 is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be

Open solution

(A) 2.0×10^{11} (B) 4.0×10^{12} (C) 1.0×10^{2} (D) 1.0×10^{10}

Q71. AIPMT 2003

The e.m.f. of a Daniell cell at 298 K is E_1.

Open solution

$$Zn \left| ZnSO_4 \right. \left| \begin{matrix} \\ (0.01\ M) \end{matrix} \right\| \left. \begin{matrix} CuSO_4 \\ (1.0\ M) \end{matrix} \right| Cu$$

When the concentration of $ZnSO_4$ is 1.0 M and that of $CuSO_4$ is 0.01 M, the e.m.f. changed to E_2.
What is the relationship between E_1 and E_2?

(A) $E_1 > E_2$ (B) $E_1 < E_2$ (C) $E_1 = E_2$ (D) $E_2 = 0 \neq E_1$

Q72. AIPMT 2003

On the basis of the information available from the reaction,

Open solution

$4/3Al + O_2 \rightarrow 2/3Al_2O_3, \quad \Delta G = -827\ kJ\ mol^{-1}$ of O_2,

the minimum e.m.f. required to carry out an electrolysis of Al_2O_3 is
($F = 96500\ C\ mol^{-1}$)

(A) 2.14 V (B) 4.28 V (C) 6.42 V (D) 8.56 V

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Q73. AIPMT 2002

In electrolysis of NaCl when Pt electrode is taken then H_2 is liberated at cathode while with Hg cathode it forms sodium amalgam

(A) Hg is more inert than Pt

(B) More voltage is required to reduce H^+ at Hg than at Pt

(C) Na is dissolved in Hg while it does not dissolve in Pt

(D) Conc. of H^+ ions is larger when Pt electrode is taken.

Open solution

Q74. AIPMT 2001

Standard electrode potentials are
Fe^{2+}/Fe [$E^o = -0.44$] and
Fe^{3+}/Fe^{2+}[$E^o = 0.77$];
If Fe^{2+}, Fe^{3+} and Fe blocks are kept together, then

(A) Fe^{3+} increases

(B) Fe^{3+} decreases

(C) Fe^{2+}/Fe^{3+} remains unchanged

(D) Fe^{2+} decreases.

Open solution

Q75. AIPMT 2000

Cell reaction is spontaneous when

(A) ΔG^o is negative

(B) ΔG^o is positive

(C) ΔE^o_{red} is positive

(D) ΔE^o_{red} is negative

Open solution

Q76. AIPMT 2000

Equivalent conductances of Ba^{2+} and Cl^- ions are 127 and 76 $ohm^{-1}\ cm^{-1}\ eq^{-1}$ respectively. Equivalent conductance of $BaCl_2$ at infinite dilution is

(A) 139.5 (B) 101.5 (C) 203 (D) 279

Open solution

Q77. AIPMT 2000

For the disproportionation of copper
$2Cu^+ \rightarrow Cu^{2+} + Cu$, E^o is
(Given E^o for Cu^{2+}/Cu is 0.34 V and
E^o for Cu^{2+}/Cu^+ is 0.15 V.)

(A) 0.49 V (B) − 0.19 V (C) 0.38 V (D) −0.38 V

Open solution

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Q78. AIPMT 1999

The specific conductance of a $0.1\ \mathrm{N\ KCl}$ solution at $23^\circ\mathrm{C}$ is $0.012\ \mathrm{ohm^{-1}\ cm^{-1}}$. The resistance of cell containing the solution at the same temperature was found to be $55\ \mathrm{ohm}$. The cell constant will be

Open solution

(A) $0.142\ \mathrm{cm^{-1}}$ (B) $0.66\ \mathrm{cm^{-1}}$ (C) $0.918\ \mathrm{cm^{-1}}$ (D) $1.12\ \mathrm{cm^{-1}}$

Q79. AIPMT 1998

For the cell reaction

$\mathrm{Cu^{2+}}(C_1\ \mathrm{aq}) + \mathrm{Zn(s)} \rightleftharpoons \mathrm{Zn^{2+}}(C_2\ \mathrm{aq}) + \mathrm{Cu(s)}$

of an electrochemical cell, the change in free energy (ΔG) at a given temperature is a function of

Open solution

(A) $\ln(C_1)$ (B) $\ln(C_2/C_1)$ (C) $\ln(C_2)$ (D) $\ln(C_1 + C_2)$

Q80. AIPMT 1998

Without losing its concentration $\mathrm{ZnCl_2}$ solution cannot be kept in contact with

Open solution

(A) Au (B) Al (C) Pb (D) Ag

Q81. AIPMT 1997

E° for the cell, $\mathrm{Zn\,|\,Zn^{2+}_{(aq)}\,||\,Cu^{2+}_{(aq)}\,|\,Cu}$ is $1.10\ \mathrm{V}$ at $25\ ^\circ\mathrm{C}$, the equilibrium constant for the reaction $\mathrm{Zn + Cu^{2+}_{(aq)} \rightleftharpoons Cu + Zn^{2+}_{(aq)}}$ is the order of

Open solution

(A) 10^{+18} (B) 10^{+17} (C) 10^{-28} (D) 10^{-37}

Q82. AIPMT 1997

The molar conductances of NaCl, HCl and $\mathrm{CH_3COONa}$ at infinite dilution are 126.45, 426.16 and $91\ \mathrm{ohm^{-1}\,cm^2\,mol^{-1}}$ respectively. The molar conductance of $\mathrm{CH_3COOH}$ at infinite dilution (Λ_m^∞) is

Open solution

(A) $698.28\ \mathrm{ohm^{-1}\,cm^2\,mol^{-1}}$ (B) $540.48\ \mathrm{ohm^{-1}\,cm^2\,mol^{-1}}$
(C) $201.28\ \mathrm{ohm^{-1}\,cm^2\,mol^{-1}}$ (D) $390.71\ \mathrm{ohm^{-1}\,cm^2\,mol^{-1}}$

Q83. AIPMT 1996

A 5 ampere current is passed through a solution of zinc sulphate for 40 minutes. The amount of zinc deposited at the cathode is

Open solution

(A) $0.4065\ \mathrm{g}$ (B) $65.04\ \mathrm{g}$ (C) $40.65\ \mathrm{g}$ (D) $4.065\ \mathrm{g}$

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Q84. AIPMT 1996

Reduction potential for the following half-cell reactions are

$\mathrm{Zn} = \mathrm{Zn}^{2+} + 2e^-;\ E^\circ = +0.76\ \mathrm{V};$

$\mathrm{Fe} = \mathrm{Fe}^{2+} + 2e^-;\ E^\circ = +0.44\ \mathrm{V}.$

The EMF for the cell reaction

$\mathrm{Fe}^{2+} + \mathrm{Zn} \rightarrow \mathrm{Zn}^{2+} + \mathrm{Fe}$ will be

(A) $-0.32\ \mathrm{V}$ (B) $+1.20\ \mathrm{V}$ (C) $-1.20\ \mathrm{V}$ (D) $+0.32\ \mathrm{V}$

Open solution

Q85. AIPMT 1995

An electrochemical cell is set up as :
$\mathrm{Pt};\ \mathrm{H}_2\ (1\ \mathrm{atm})\ |\ \mathrm{HCl}\ (0.1\ \mathrm{M})\ ||\ \mathrm{CH_3COOH}\ (0.1\ \mathrm{M})\ |\ \mathrm{H}_2\ (1\ \mathrm{atm});\ \mathrm{Pt}$. The e.m.f. of this cell will not be zero, because

(A) acids used in two compartments are different

(B) e.m.f. depends on molarities of acids used

(C) the temperature is constant

(D) pH of 0.1 M HCl and 0.1 M $\mathrm{CH_3COOH}$ is not same

Open solution

Q86. AIPMT 1994

Standard reduction potentials at $25\ ^\circ\mathrm{C}$ of $\mathrm{Li}^+|\mathrm{Li}$, $\mathrm{Ba}^{2+}|\mathrm{Ba}$, $\mathrm{Na}^+|\mathrm{Na}$ and $\mathrm{Mg}^{2+}|\mathrm{Mg}$ are -3.05, -2.90, -2.71 and -2.37 volt respectively. Which one of the following is the strongest oxidising agent?

(A) Ba^{2+} (B) Mg^{2+} (C) Na^+ (D) Li^+

Open solution

Q87. AIPMT 1994

To protect iron against corrosion, the most durable metal plating on it, is

(A) copper plating (B) zinc plating (C) nickel plating (D) tin plating.

Open solution

Q88. AIPMT 1993

When $\mathrm{CuSO_4}$ is electrolysed using platinum electrodes,

(A) copper is liberated at cathode, sulphur at anode

(B) copper is liberated at cathode, oxygen at anode

(C) sulphur is liberated at cathode, oxygen at anode

(D) oxygen is liberated at cathode, copper at anode.

Open solution

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Q89. AIPMT 1992

On electrolysis of dilute sulphuric acid using platinum electrodes, the product obtained at the anode will be

Open solution

(A) hydrogen

(B) oxygen

(C) hydrogen sulphide

(D) sulphur dioxide

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Answer Key

1. **C**	2. **B**	3. **B**	4. **D**	5. **B**
6. **A**	7. **A**	8. **B**	9. **B**	10. **D**
11. **A**	12. **B**	13. **D**	14. **D**	15. **C**
16. **D**	17. **A**	18. **A**	19. **C**	20. **D**
21. **C**	22. **D**	23. **A**	24. **D**	25. **A**
26. **C**	27. **B**	28. **C**	29. **D**	30. **B**
31. **B**	32. **B**	33. **A**	34. **C**	35. **D**
36. **C**	37. **A**	38. **C**	39. **C**	40. **D**
41. **D**	42. **B**	43. **A**	44. **C**	45. **D**
46. **A**	47. **D**	48. **C**	49. **D**	50. **C**
51. **C**	52. **B**	53. **B**	54. **B**	55. **D**
56. **A**	57. **A**	58. **D**	59. **A**	60. **D**
61. **A**	62. **A**	63. **A**	64. **C**	65. **B**
66. **A**	67. **D**	68. **D**	69. **C**	70. **D**
71. **A**	72. **A**	73. **B**	74. **B**	75. **A**
76. **A**	77. **C**	78. **B**	79. **B**	80. **B**
81. *Scan QR*	82. **D**	83. **D**	84. **D**	85. **D**
86. **B**	87. **C**	88. **B**	89. **B**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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